Wolters Kluwer

CCH® Tagetik | ACADEMY



CCH® Tagetik - OData Integration

When you have to be right

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General notes

Please read this page carefully before jumping right into the technical specifications.

Purpose of this document

This document is addressed to CCH® Tagetik implementers. It provides the reader with the guidelines to setup a connection that allows third-party software to consume CCH® Tagetik data, exposed by RESTful API with standard OData.

An additional paragraph explains how Tagetik can consume data from external sources exposed by the standard OData.

Version of this document

All information and settings in the present document are related to CCH® Tagetik version 5.3SP15.

Glossary

- **RESTful API**: application program interface (API) that uses HTTP requests to access and use data
- **OData**: Open Data Protocol used for building and consuming RESTful APIs
- **OASIS standard**: It is a Protocol developed by the non-profit Organization for the Advancement of Structured Information Standards
- **JSON file**: A JSON file is a file that stores simple data structures and objects in JavaScript Object Notation (JSON) format, which is a standard data interchange format. It is primarily used for transmitting data between a web application and a server
- **SAML**: Security Assertion Markup Language (SAML) is an open standard that allows identity providers (IdP) to pass authorization credentials to service providers (SP)
- **SCIM**: System for Cross-domain Identity Management (SCIM) is a standard for automating the exchange of user identity information between identity domains, or IT systems
- **Swagger** : Swagger is an Interface Description Language for describing RESTful APIs expressed using JSON. In CCH® Tagetik, it is used to manage the data model

How to expose CCH® Tagetik via OData

This chapter shows how to consume OData exposed by CCH® Tagetik.

Typical business case in CCH® Tagetik project

Data stored in CCH® Tagetik can be consumed with CCH® Tagetik standard reporting or be analyzed using a third-party BI tool.

When CCH® Tagetik is deployed on cloud, it is necessary to expose the data by means of RESTful API and OData service to allow third-party BI tools to read it and display it.

RESTful API and OData service

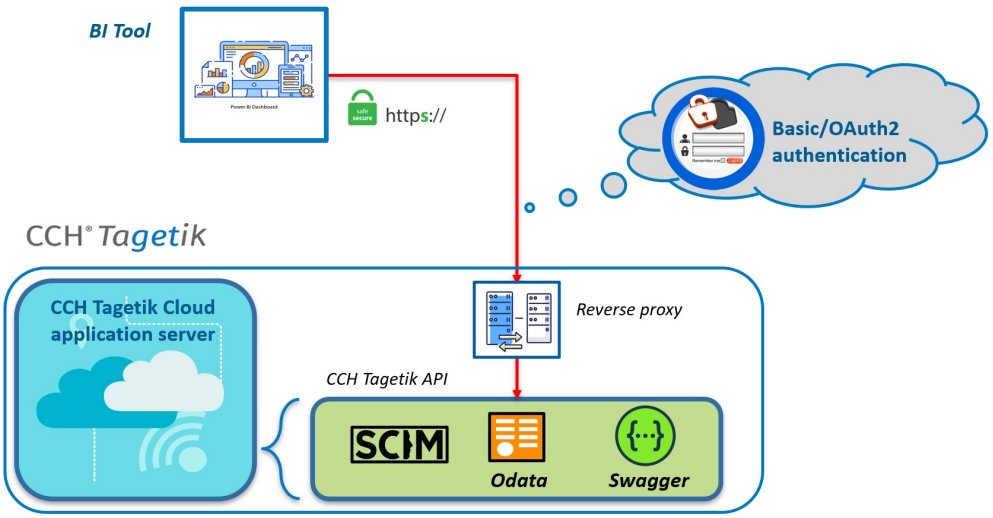
A RESTful API is an architectural style for an application program interface (API) that uses HTTP requests to access and use data. OData (Open Data Protocol) is an ISO/IEC approved, OASIS standard that defines a set of best practices for building and consuming RESTful APIs.

CCH® Tagetik exposes its data by RESTful APIs using in OData protocol. At present, the OData service used is version 4. It means that the CCH® Tagetik APIs generate a JSON file that the browser translate into an XML.

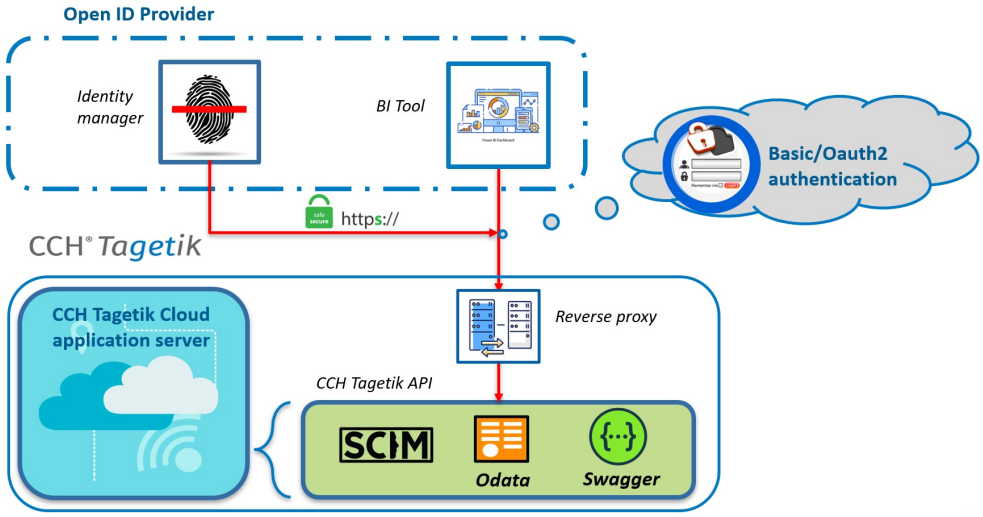
Why is OData service the solution?

CCH® Tagetik allows to expose its data by RESTful APIs using the OData protocol. OData eases the access of data from external connectors without a direct connection to the database. Consumers can access CCH® Tagetik data directly via HTTPs requests.

Typical frameworks



The image above represents the most typical connection's framework . The BI tool is standalone and it is not installed in an External ID provider. CCH® Tagetik data on cloud is exposed by the different RESTful API. SCIM represents information for user provisioning, Swagger is a standard for metadata whereas the OData standard provides the real Data of CCH® Tagetik. The BI tool pulls its requests to Tagetik, and the reverse proxy allows the whitelisted IP to consume the data exposed by API. The user will be authenticated by a Basic authentication or by an OAuth2 authentication.



In this second framework the BI tool is installed on an Open ID provider (could be for example Azure). Using this solution, the preferred user authentication is Oauth2 because it provides a stronger security for the data. However, the use of basic authentication is allowed too. The Identity manager of the open ID provider could also consume SCIM for user provisioning.

In both solutions all data flows are encrypted using HTTPS protocol.

CCH® Tagetik setup

Below the most important settings of CCH® Tagetik.

Tagetik Application Server

The System Administrator must activate REST APIs via Repository in System Administration > Security > Checkbox Enable public REST APIs

Firewalls and Whitelists

All IP Public addresses that need to connect to CCH® Tagetik and to OData must be whitelisted due to proxy.

Implementer has to open an INCIDENT on CCH® Tagetik support portal to Cloud team that includes IP Public addresses to be whitelisted. Cloud team will manage to update the whitelist and to setup properly the file standalone.xml. For this setup it is necessary provide also the URL used to access the Tagetik application

IPs to whitelist in case of Power BI cloud-based or standalone

When PowerBI is installed on a local computer, the list of IPs to be whitelisted can be univoquely identified. Instead, when deployed on the cloud, the IP depends on the region and on the service required; if it is deployed on Microsoft Azure, Microsoft itself provides a range of available IP addresses, among which the Customer IT have to choose the ones to use. Here is the link to use to retrieve the MS Azure IPs: https://www.microsoft.com/en- us/download/details.aspx?id=56519 . If PowerBI is deployed using a different provider, Customer IT should contact the provider to request the available IPs.

Authentication

Implementer has to choose the kind of user authentication to use in the framework. There are 2 kinds of Authentication systems to access API services:

- Basic
- oAuth2

Basic Authentication

It is the preferred solution when installation does not need an external OpenID provider where to register one or more client applications.

Users involved in the connection must be defined by CCH® Tagetik system administrator with these properties:

- To be defined as Local Tagetik user
- The password must be specified during user definition
- The option “Password to change” must be disabled
- To be able to access OData Stream fromCCH® Tagetik User should be assigned to a custom role enabled to Tagetik Function 0004077

It is not possible to use credential for external identity like Single-sign-on or SAML

The credential (Username and password) in phase of authentication will be the same defined for the Tagetik user

OAUTH2 Authentication

This is the preferred kind of authentication to use when connector/BI tool is installed on an external OpenID provider where to register one or more client applications. These are the steps to setup this kind of configuration:

- Create one or more Local Users in CCH® Tagetik Repository and give them the proper roles. They have to be flagged as Reference Users. These users can be marked as “disabled” since they won’t access directly to the application.
- To be able to access OData Stream from Tagetik Reference User should be assigned to a custom role enabled to Tagetik Function 0004077
- Implementer has to open an INCIDENT on CCH® Tagetik support portal to Cloud team or IT in which the implementer will ask to Create configuration file with the OpenID properties to be stored on the application server’s local filesystem. These properties have the information CCH Tagetik will use to communicate with the OpenID provider.
- Setup of Users:
    - CCH® Tagetik system administrator has to define one or more Local Users in CCH® Tagetik Repository and give them the proper roles. They have to be flagged as Reference Users. Each Reference user represents a different scope in the Client Identity Provider. These users can be marked as “disabled” since they won’t access directly to the application.
    - Customer IT has to configure, on OpenID provider, an OAuth 2.0 token with Client Credential Flow. Each Scope must be coded as the related Reference user defined in Tagetik. This token should contain one or more usernames of the Reference Users created above as Scopes.

OAuth2 is not supported directly by PowerBI. In case, customer should create a specific driver to connect API to PowerBI. Tagetik does not support this implementation

Responsibilities

Implementing OData connection is a joint effort of different actors involved:

- Implementer
    - Supervise the project and assign the different responsibilities. In some case the implementer has the duties of System administrator
- System administrator:
    - REST APIs activation
    - Setup users and roles
- Cloud team:

    - Setup of proxy's whitelist
    - Setup of file standalone.xml
    - In case of OAuth2 authentication: Connection/Authentication with Open IP provider
- Customer IT:
    - In case of OAuth2 authentication configuration of Open IP provider
    - Provide the list of the IPs to be whitelisted

OData Documentation and tips

Please be aware of this:

- Information about RESTful API (OData, SCIM, Swagger) can be retrieved from CCH® Tagetik online user manual: RestAPI section
- Data present in Analytic Information Hub including Dataset, Virtual dataset and Datasource can be exposed directly by OData.
- OData expose all data present in dataset without any restriction (user rights)
- Financial Workspace data cannot be exposed directly. They can be exposed through AIH Datasource that will set them available for external connectors
- Information about exposure of Analytic Information Hub data can be retrieved from CCH® Tagetik online user manual: Analytic Information Hub > AIH exposure through RESTful APIs section

Implementer skills required

- Tagetik Repository: basic knowledge
- Analytic Information Hub: advanced knowledge
- Network Administration: basic knowledge

TODO List: Basic Authentication

This is a recap of the action required to setup the connection via OData

- Activate REST API from Repository
- Open an INCIDENT on CCH® Tagetik support portal to Cloud team that includes IP Public addresses to be whitelisted and URL used to access the Tagetik application
- Create CCH® Tagetik users and roles

TODO List: OAuth2 Authentication

This is a recap of the action required to setup the connection via OData

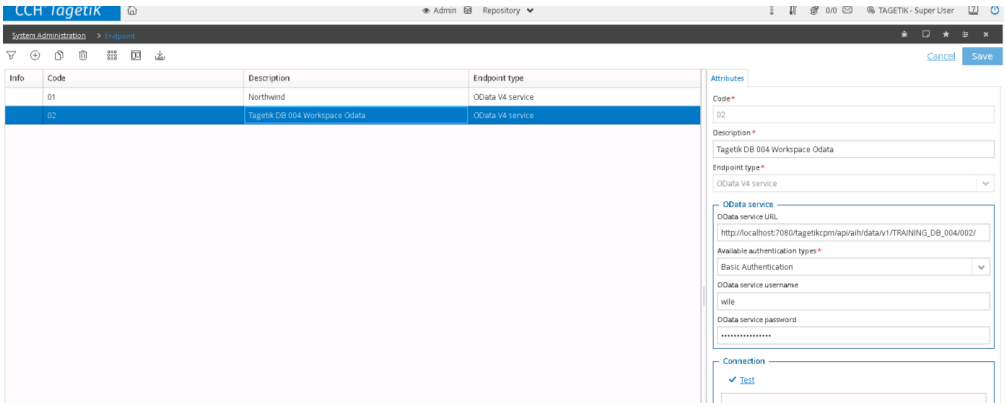
- Activate REST API from Repository
- Open an INCIDENT on CCH® Tagetik support portal to Cloud team that includes IP Public addresses to be whitelisted and URL used to access the Tagetik application
- Create CCH® Tagetik users and roles
- Open an INCIDENT on CCH® Tagetik support portal to Cloud team to create configuration file with the OpenID properties to be stored on the application server's local filesystem
- Configure an OAuth 2.0 token with Client Credential Flow on OpenID provider (Customer IT responsibility)
- Create a specific driver to connect API to PowerBI

How to load OData in CCH® Tagetik

CCH® Tagetik can consume data exposed from external sources via OData service using as landing of the data the AIH or the ETL.

Endpoint

First step is the creation of an Endpoint that represents the URL of the source that exposes OData. It can be defined by System administrator from Repository > Endpoint > New Element.



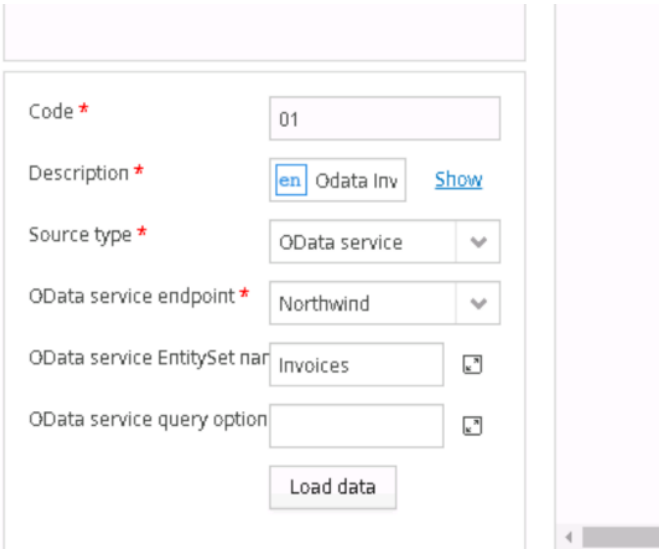
Endpoint type has to be set as OData V4 service or Odata V2 Service depending on the Version of OData used by the source. CCH® Tagetik supports Version 2 (SAP exposes its data in Version 2) or Version 4. The 2 versions differ in their syntax. After Endpoint selection these are the other fields to setup:

- OData service URL
- Available authentication type: None, Basic, OAuth2
- OData service username
- OData service password

Loading OData in AIH

Data exposed by OData can be loaded in a dataset defining a Data source mapping

The user has to set-up the Attributes.



Data source Mapping

- Source type: OData service
- OData service endpoint: Choose the related endpoint among the one available in the Repository
- OData service EntitySet name : name of entities collection (the EntytySet described in the Entity data model of the service's root)
- OData service query options: Defines specific Filter (optional)

Clicking on "Load Data" the system retrieves all the fields held in the EntitySet selected and filtered optionally by the query option. Then a mapping between Dataset's fields and EntitySet's field has to be defined.

Execution can be performed from workflow or scheduled as a standard datasource

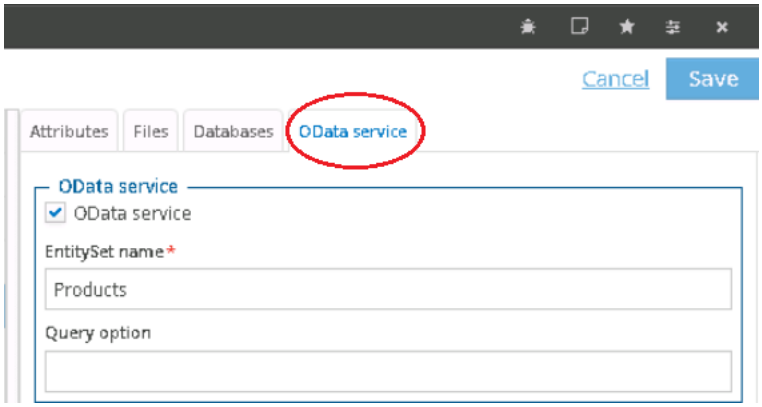
Loading OData in ETL

Beside the option to use a Data source Mapping within an Analytical Workspace Dataset, it is possible to use Odata as source of an ETL extraction.

While defining a data source in ETL Domain, OData service is the tab to select.

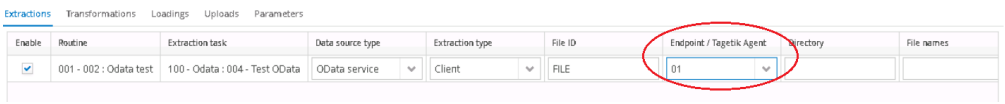
The checkbox OData service has to be ticked, as well the following fields has to be filled:

- OData service EntitySet name: name of entities collection (the EntytySet described in the Entity data model of the service's root)
- OData service query options: Defines specific Filter (optional)



The fields of the Entity set have to be mapped with Staging table fields.

While running a routine, the Endpoint from which consumes the OData has to be chosen manually as in the figure below



If user decides to include the routine that extracts OData in a Job, the Endpoint can be preconfigured in the list of Job's extractions as in figure below or chosen at runtime

